

A Twenty-First Century Cancer Epidemic Caused by Obesity: The Involvement of Insulin, Diabetes, and Insulin-Like Growth Factors

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Obesity has reached epidemic proportions in the developed world. The progression from obesity to diabetes mellitus type 2, via metabolic syndrome, is recognised, and the significant associated increase in the risk of major human cancers acknowledged. We review the molecular basis of the involvement of morbidly high concentrations of endogenous or therapeutic insulin and of insulin-like growth factors in the progression from obesity to diabetes and finally to cancer. Epidemiological and biochemical studies establish the role of insulin and hyperinsulinaemia in cancer risk and progression. Insulin-like growth factors, IGF-1 and IGF-2, secreted by visceral or mammary adipose tissue have significant paracrine and endocrine effects. These effects can be exacerbated by increased steroid hormone production. Structural studies elucidate how each of the three ligands, insulin, IGF-1, and IGF-2, interacts differently with isoforms A and B of the insulin receptor and with type I IGF receptor and explain how these protagonists contribute to diabetes-associated cancer. The above should inform appropriate treatment of cancers that arise in obese individuals and in those with diabetes mellitus type 2. Novel drugs that target the insulin and insulin-like growth factor signal transduction pathways are in clinical trial and should be effective if appropriate biomarker-informed patient stratification is implemented.